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and is covered by three cranial meninges. The outer thick layer is **Duramater** which lines the inner surface of the cranial cavity; the median thin layer is **Arachnoid mater** which is separated from the duramater by a narrow **subdural space.** The innermost layer is **Piamater** which is closely adhered to the brain but separated from the arachnoid mater by the **subarachnoid space**. The brain is divided into three major regions: Forebrain, Midbrain and Hindbrain.

Human brain is formed of a large number of parts like cerebrum, thalamus, hypothalamus, pons, cerebellum and medulla oblongata. Each part performs some specialized function and all the parts are essential for the survival of a person. Discuss the following statements :

a) Thalami are called relay centres of the brain.

b) Damage to medulla may cause the death of organism.

Fore Brain

It comprises the following regions: **Cerebrum** and **Diencephalon**. Cerebrum is the 'seat of intelligence' and forms the major part of the brain. The cerebrum consists of an outer cortex, inner medulla and **basal nuclei.** The superficial region of the cerebrum is called **cerebral cortex**, which looks grey due to the presence of unmyelinated nerve cells. Cerebral cortex consists of neuronal cell body, dendrites, associated glial and blood vessels. The surface of the cerebrum shows many convolutions (folds) and grooves. The folds are called **gyri** (singular gyrus); the shallow grooves between the gyri are called **sulci** (singular sulcus) and deep grooves are called fissures. These sulci and gyri increase the surface area of the cerebral cortex. Several sulci divide the cerebrum into eight lobes: a pair of **frontals, parietals, temporals and occipital lobes** (Figure10.7 & Table 10.2).

A median longitudinal fissure divides the cerebrum longitudinally into two cerebral hemispheres (Figure 10.7). A transverse fissure separates the cerebral hemispheres from the cerebellum. The

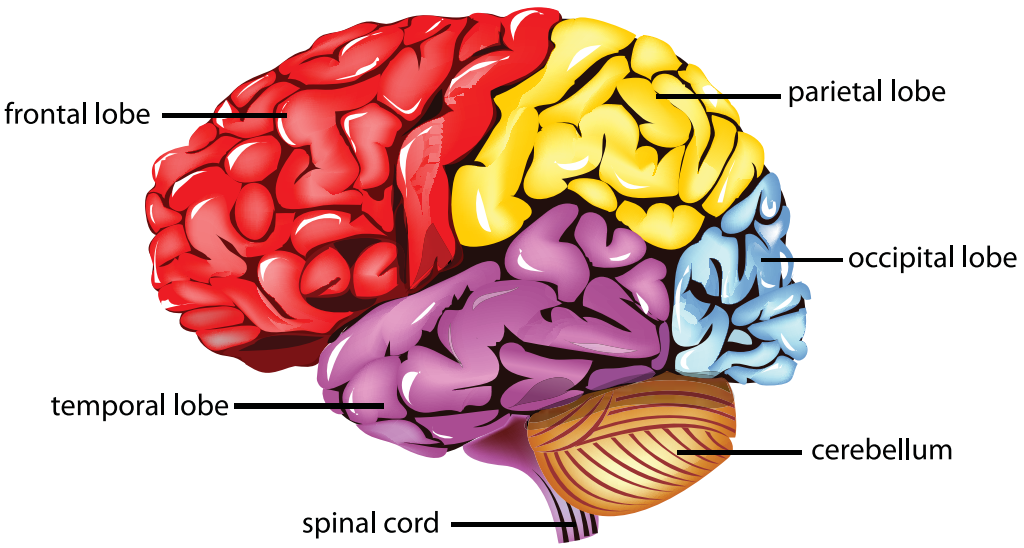


Figure 10.7 Lobes of Cerebral hemisphere